Chapter 2

Microscopy

Learning Objectives

After studying this chapter the students will be able to,

- *Identify the principle components of Phase Contrast, Fluorescence and Electron Microscope.*
- *Understand the optics in different light microscope and image formation mechanism.*
- *Know the principle, working mechanism of Phase Contrast, Fluorescent Microscope and Electron Microscope.*
- *Differentiate Light and Electron Microscope.*
- *Appreciate the applications of Phase Contrast, Fluorescence and Electron Microscopes.*

Chapter Outline

2.1 Phase Contrast Microscope

2.2 Fluorescence Microscope

2.3 Electron Microscope

Microscopes are specialized optical instruments designed to produce magnified visual or photographic images of objects or specimens that are too small to be seen with naked eye. Today, more sophisticated compound light microscopes are routinely used in microbiology laboratories. In the previous year, we have learnt about light microscopes that includes bright field and Dark-field microscopes. This year we are going to learn about other types of light microscopes such as phase contrast and fluorescence microscopes. Yet another well advanced microscope which uses electron as source rather than light – the electron microscope is also discussed in detail in this chapter.

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2.1 Phase Contrast Microscope

Frits Zernike a Dutch Physicist invented the Phase Contrast Microscope and was awarded Nobel Prize in 1953. It is the microscope which allows the observation of living cell. This microscopy uses special optical components to exploit fine differences in the refractive indices of water and cytoplasmic components of living cells to produce contrast.

2.1.1 Principle

The phase contrast microscopy is based on the principle that small phase changes in the light rays, induced by differences in the thickness and refractive index of the different parts of an object, can be transformed into

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